19 Coffee: Forecasting Futures

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Goal & Hypotheses

We evaluate six forecasting models to determine which best predicts coffee futures prices over a 63-trading-day (quarterly) horizon, and to investigate whether historical price patterns contain information exploitable by complex models. We hope to predict when 19 Coffee should purchase futures.

- **H_1:** Complex, pretrained models will outperform simple statistical baselines.
- **H_2:** Model rankings will stabilize as the number of forecasts increases.

What is 19 Coffee Company?

- Small-owned Pittsburgh coffee roaster
- Source beans grown without pesticides or herbicides, under natural canopies
  - All directly traded, relationship, fair trade certified, organic certified, and shade grown coffee
- Partners:
  - Building New Hope: support livelihoods of Central Americans through education
  - Firsthand Cooperative: uses fair share agreement, connecting small farms with consumers

Why Forecasting is Hard

Coffee futures prices are driven by many interacting forces, none of which appear directly in historical price data:

- **Exogenous:** Tariffs, weather, trade policy, global market shocks
- **Endogenous:** Production volume, demand, inventory levels, operational costs

If markets efficiently absorb this information, historical prices alone may contain no exploitable patterns.

What is a Futures Contract?

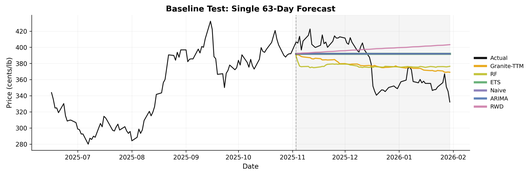
Futures Contract: a standardized agreement to buy or sell a commodity at a predetermined price on a specific expiration date

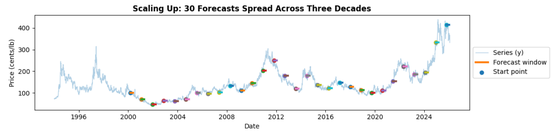
Models Evaluated

Using 32 years of daily Coffee C futures from ICE (8,104 trading days), we evaluated six models:

Category	Model	Description
Simple Baselines	Naïve	Last observed price repeated forward
	RWD	Last price plus average daily drift
Classical Statistical	ETS	Exponential smoothing; weights recent prices higher
	ARIMA	Automatic order selection
Machine Learning	Random Forest	400-tree ensemble regression model; recursive forecast
Foundation Model	Granite TTM	IBM pretrained transformer; zero-shot

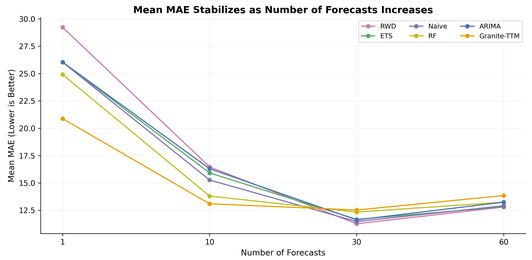
All models received the same 1,536-day context window and predicted 63 days ahead. No model saw test data during fitting.

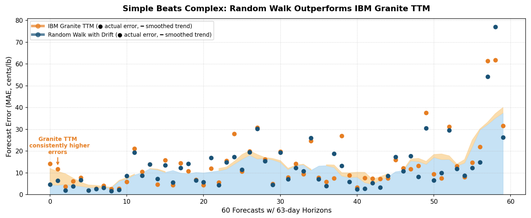




We tested at 1, 10, 30, and 60 evenly spaced forecasts. At 60 forecasts, no forecast periods overlap, ensuring independent evaluation data.

Results: Model Comparison





Simple baselines (RWD, Naïve) matched or outperformed all complex models. Granite TTM and Random Forest performed 3–8% worse despite vastly greater complexity.

Next Steps

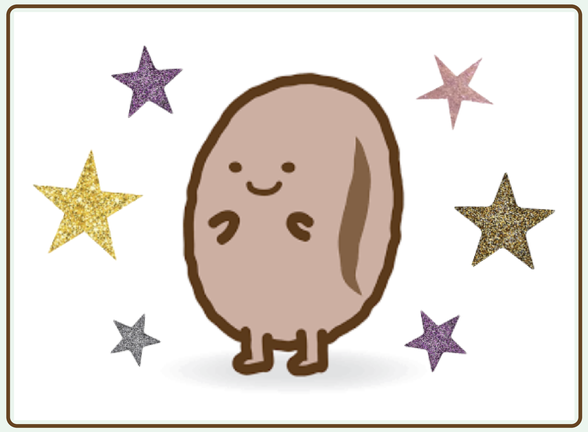
By incorporating exogenous data sets, such as tariffs and weather, into a SARIMA model, we aim to develop a SARIMAX model to obtain more accurate predictions for 19 Coffee.

Key Findings

- **H_1 rejected**: **Efficient Market Hypothesis** implies traders continuously exploit predictable patterns until those patterns vanish, leaving no learnable signal for complex models to capture.
- **H_2 confirmed:** Rankings were unstable at low forecast counts but stabilized by 30–60 forecasts, demonstrating that sufficient evaluation windows are required for reliable model comparison.
- **Simple beats complex:** RWD outperformed IBM Granite TTM despite requiring no training data or GPU compute.

Technology & Sources

- **Programming:** Python (NumPy, Pandas, SciPy)
- **Statistical:** `statsforecast` (AutoARIMA, AutoETS, Naïve, Random Walk with Drift)
- **Machine Learning:** scikit-learn (Random Forest)
- **Deep Learning:** IBM Granite TTM (PyTorch, CUDA Acceleration)
- **Data Source:** ICE Coffee C Futures — Daily Nearby Close Prices (1993–2025)

Acknowledgements

- Department of Mathematics, University of Pittsburgh
- Dr. Jeffrey Wheeler
- 19 Coffee Company